ModelIt!

Teacher's Guide

Microplastics in the Food Chain: The Invisible Invasion

Modeling Bioaccumulation and Trophic Transfer of Persistent Pollutants

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-LS2-6, HS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-6, HS-ESS3-3

Evaluate claims that complex interactions in ecosystems maintain relatively consistent conditions. Create a computational simulation of a natural system.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze how microplastic concentrations change across trophic levels and over time to identify patterns of bioaccumulation.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy Transfer** — Matter cycles through ecosystems, but persistent pollutants like microplastics accumulate rather than cycle, concentrating at higher trophic levels.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students model how small-scale plastic inputs create large-scale concentration effects through biomagnification across trophic levels.

CAST Testing Alignment

MULTIPLE CHOICE:

Which strategy would MOST reduce human microplastic exposure from seafood: (a) ocean cleanup, (b) reducing new plastic production, (c) cooking fish differently, (d) eating farmed fish?

CONSTRUCTED RESPONSE:

Using your model, explain why a 50 percent reduction in plastic input today would take decades to produce a measurable reduction in seafood microplastic levels.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

An estimated 8 million metric tons of plastic enter the oceans every year. This plastic does not decompose -- it breaks into smaller and smaller pieces that enter food chains and accumulate in organisms, including humans.

Sources of Microplastics

Microplastics come from degradation of larger plastic waste, synthetic clothing fibers released in laundry, microbeads in personal care products, and tire wear. A single wash of synthetic clothing releases 700,000 microfibers.

Bioaccumulation Mechanics

Organisms ingest microplastics by mistaking them for food or passively filtering contaminated water. Because plastic cannot be digested or excreted efficiently, it accumulates in tissues over the organism lifetime.

Trophic Amplification

A small fish eats thousands of contaminated plankton. A larger fish eats hundreds of small fish. At each level, the predator inherits the accumulated plastic of all its prey. By the time you reach a top predator, concentrations can be millions of times higher than in the water.

Human Health Implications

The average person ingests approximately 5 grams of plastic per week -- roughly the weight of a credit card. Microplastics have been found in human blood, lungs, and placenta. Health effects are still being studied but include inflammation, endocrine disruption, and cellular damage.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify plastic input, microplastic concentration, bioaccumulation, trophic transfer, organism health, decomposition rate, and human exposure as system components.

E **Establish Relationships**
Students map the one-way accumulation pathway from input through trophic levels to human exposure, identifying amplification at each step.

V **Visualize & Model**
Students build a model showing how bioaccumulation and biomagnification create disproportionate contamination at higher trophic levels.

E **Evaluate Outcomes**
Students run current trajectory, reduction, and trophic amplification scenarios to understand timescales and intervention effectiveness.

R **Revise & Extend**
Students add chemical leaching, wastewater treatment, or bioremediation to explore additional intervention strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You eat a credit card worth of plastic every week. How does it get from a trash can to your dinner plate?"*

Watch For

Show the statistic. Display microplastic images.

Slide 2: Learning Objectives

3 min

Say: *"Today we trace the invisible journey of plastic through the entire food chain to your body."*

Watch For

Pre-teach bioaccumulation and biomagnification.

Slide 3: Big Question

3 min

Say: *"How does plastic from the middle of a continent end up in ocean fish thousands of miles away?"*

Watch For

Quick-write: Trace the path of a plastic bag from trash to table.

Slide 4: LEVER Framework

2 min

Say: *"Seven components. A one-way ratchet. The plastic never goes away."*

Watch For

Preview why this system only accumulates.

Slide 5: Activity 1: Components

8 min

Say: *"What can humans control? What is beyond our current ability to change?"*

Watch For

Sort components. Discuss why decomposition rate is effectively zero.

Slide 6: Activity 2: Connect

10 min

Say: *"At each level of the food chain, concentration INCREASES. Why?"*

Watch For

Map the amplification pathway. Calculate concentration ratios.

Slide 7: Activity 3: Simulate

10 min

Say: *"Even if we stopped ALL plastic today, how long until the ocean is clean?"*

Watch For

Students discover the persistence problem through scenarios.

Slide 8: Discoveries

5 min

Say: *"This is a problem that gets worse even when we start fixing it. That is the nature of persistence."*

Watch For

Discuss the timescale mismatch between human action and environmental response.

Slide 9: STEM Challenge

12 min

Say: *"A fishing industry is at stake. Consumers are worried. Design a system that tracks and reduces the threat."*

Watch For

Groups design monitoring and intervention systems.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Plastic Input Rate, Decomposition Rate
- **INTERNAL:** Microplastic Concentration, Bioaccumulation Factor, Trophic Transfer, Marine Organism Health, Human Exposure Level

Relationships

- **Plastic Input Rate to Microplastic Concentration:** POSITIVE (+) — More plastic entering the environment increases the concentration of microplastic particles.
- **Decomposition Rate to Microplastic Concentration:** NEGATIVE (-) — Faster decomposition would reduce concentration, but plastic decomposition is effectively zero.
- **Microplastic Concentration to Bioaccumulation Factor:** POSITIVE (+) — Higher environmental concentration leads to more ingestion and accumulation in organisms.
- **Bioaccumulation to Trophic Transfer:** POSITIVE (+) — More accumulated plastic in prey means more transferred to predators.
- **Trophic Transfer to Human Exposure:** POSITIVE (+) — Higher trophic transfer concentrates more plastic in seafood consumed by humans.
- **Bioaccumulation to Marine Organism Health:** NEGATIVE (-) — Higher plastic loads cause inflammation, organ damage, and reduced fitness.
- **Marine Organism Health to Trophic Transfer:** NEGATIVE (-) — If organisms die from contamination, fewer are available as prey, potentially reducing but not eliminating transfer.

Key Discoveries

- Microplastic concentration in the environment is essentially a one-way ratchet because decomposition is negligible compared to input
- Bioaccumulation amplifies concentration by 10-100x at each trophic level, meaning top predators carry enormously higher loads
- Reducing plastic input slows but does not reverse the problem because existing plastics persist for centuries
- Human exposure is significant and growing -- the average person ingests a credit card worth of plastic per week

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The ocean is so big that plastic gets diluted to harmless levels"	The ocean is large but plastic does not dilute -- it concentrates. Bioaccumulation means organisms collect and concentrate plastic from the water. Biomagnification means each trophic level amplifies further. A tuna contains millions of times more microplastic per gram than the ocean water it swims in.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Recycling solves the plastic problem"	Only 9 percent of all plastic ever made has been recycled. Most plastic is used once and discarded. Recycling slows input but does not address the 5+ billion metric tons already in the environment. And recycled plastic still eventually becomes microplastic.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Microplastics are too small to cause harm"	Microplastics carry toxic additives like phthalates and BPA that leach into tissue. They cause physical inflammation when lodged in organs. They have been found in human blood, lungs, liver, and placenta. The health effects of chronic low-level exposure are still being studied, but early evidence suggests inflammation, endocrine disruption, and cellular damage.